

Total No. of Questions : 10]

SEAT No. :

P3751

[5561]-150

[Total No. of Pages : 3

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(2012 Pattern) (Semester - II) (End Semester) (Elective - IV) (402050 B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Figures to the right indicate full marks.
- 2) Use of electronic pocket calculator is allowed.
- 3) Assume suitable data if necessary.

Q1) a) With neat sketch, explain p and h methods of mesh refinement in Finite Element Analysis. [6]

b) State and explain characteristics of global stiffness matrix. [4]

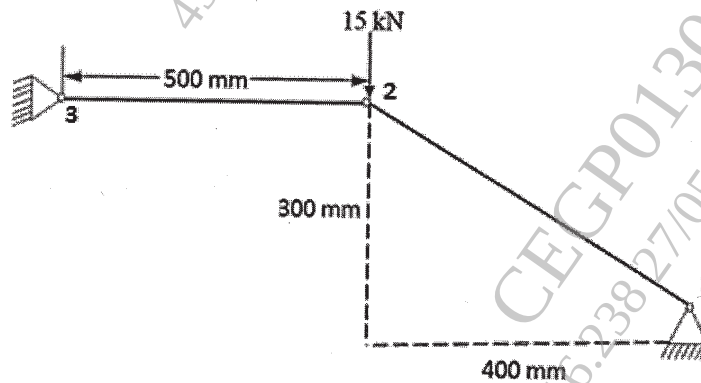
OR

Q2) a) Explain “Galerkin Weighted Residual Method” to formulate FEM equations. [6]

b) State stress-strain relations for plane stress and plane strain formulations in elasticity. [4]

Q3) a) Derive the expression for load vector for beam element due to distributed load P. [4]

b) For the two-bar truss shown in figure below, determine the displacement of node 2 and stresses in the element. All the elements have $E = 70 \text{ GPa}$ and $A = 200 \text{ mm}^2$. [6]



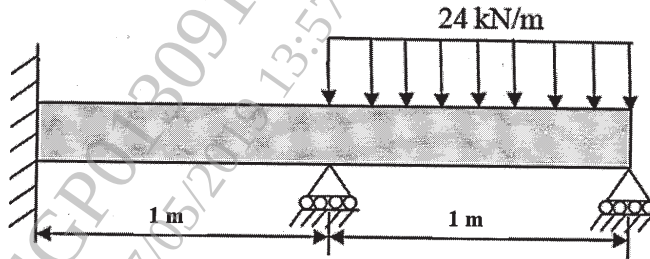
OR

P.T.O.

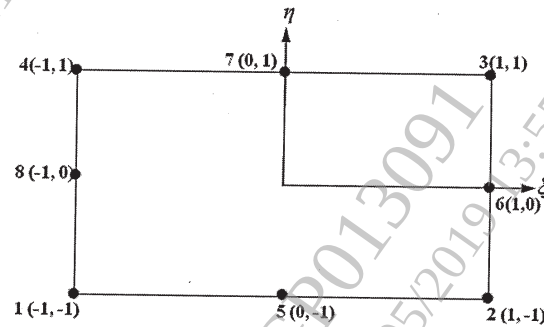
Q4) For the beam shown in figure below, determine

[10]

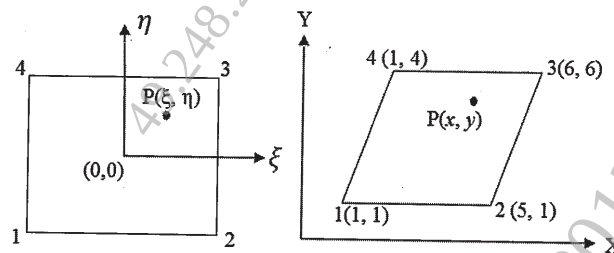
- slopes at each node and,
- vertical deflection at the mid-point of the distributed load. Take $E = 200 \text{ GPa}$ and $I = 5 \times 10^6 \text{ mm}^4$.



Q5) a) Obtain the shape functions N_1, N_2, N_3 and N_7 for following serendipity eight node rectangular element. [8]



- For four node quadrilateral element, find the x and y coordinates of point P whose location in parent element are given by $\xi = 0.5$ and $\eta = 0.5$. Also find u, v displacements of point P in X and Y directions respectively if displacement vector is $\{q\} = [0 \ 0 \ 0.20 \ 0 \ 0.15 \ 0.10 \ 0 \ 0.05]^T$. [10]



OR

Q6) a) What is full and reduced integration in Finite Element Analysis? [4]

b) Write a short note on Patch Test. [4]

- Evaluate $I = \left(\int_{y=4}^{y=6} \int_{x=-2}^{x=2} (1-x)^2 (4-y)^2 dx dy \right)$ using Gauss Quadrature method. [10]

- Q7) a)** Derive elemental stiffness matrix (conduction + convection) formulations for 1D steady state heat transfer problems. [8]
- b)** A metallic fin, with thermal conductivity $70 \text{ W/m } ^\circ\text{K}$, 1 cm radius and 5 cm long extends from a plane wall whose temperature is 140°C . Determine the temperature distribution along the fin if heat is transferred to ambient air at 20°C with heat transfer coefficient of $5 \text{ W/m}^2 \text{ } ^\circ\text{K}$. Take two elements along the fin. Assume that the tip of fin is insulated. [8]

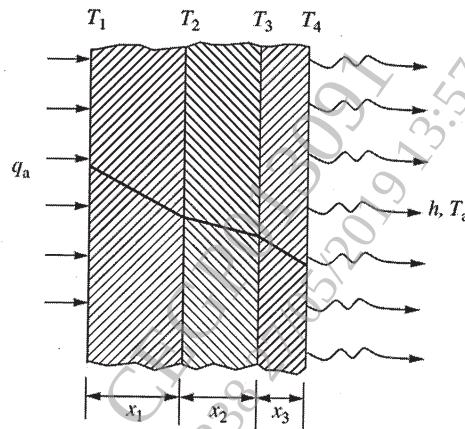
OR

- Q8)** A composite wall, with three layers of different material as shown in figure below has the following properties for the different layers: [16]

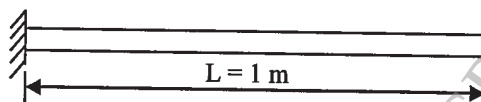
Layer-3: Gypsum, $k_3 = 0.05 \text{ W/m } ^\circ\text{C}$, $x_3 = 1 \text{ cm}$ and $q = 15 \text{ W/m}^2$

Layer-2: Fibre-glass, $k_2 = 0.0332 \text{ W/m } ^\circ\text{C}$ and $x_2 = 5 \text{ cm}$

Layer-1: Concrete, $k_1 = 1.2 \text{ W/m } ^\circ\text{C}$, $x_1 = 15 \text{ cm}$, $h = 15 \text{ W/m}^2 \text{ } ^\circ\text{C}$ and $T_a = 25^\circ\text{C}$. Calculate the temperatures T_1 , T_2 , T_3 and T_4 assuming unit area of heat flow.



- Q9)** Estimate natural frequencies of axial vibrations of bar shown in figure below, using both consistent and lumped mass matrices and compare the results. Bar is having uniform cross-section with cross-sectional area $A = 30 \times 10^{-6} \text{ m}^2$, length $L = 1 \text{ m}$, modulus of elasticity $E = 2 \times 10^{11} \text{ N/m}^2$ and density $\rho = 7800 \text{ kg/m}^3$. Model the bar by using two elements. [16]



OR

- Q10)a)** Derive consistent mass matrix for bar and truss element. [10]
- b)** Explain the significance of lumped mass matrix and consistent mass matrix. Write lumped mass matrix for bar and beam element. [6]

x x x